

1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile.

Task -1: Create Project Structure as follows:

```
OSSP/
├── .git/           # Git repository (created automatically)
├── .gitignore      # Files/folders Git should ignore
├── README.md       # Project description and instructions
├── LICENSE         # Project license (optional)
├── Makefile        # Build instructions
├── src/            # Source code
│   ├── main.c
│   ├── shell.c
│   ├── parser.c
│   ├── executor.c
│   └── builtin.c
├── include/        # Header files
│   ├── shell.h
│   ├── parser.h
│   ├── executor.h
│   └── builtin.h
├── obj/            # Object files (.o) (generated)
├── bin/            # Executable file (generated)
│   └── my_shell
├── docs/           # Documentation
│   ├── design.md
│   └── report.pdf
├── tests/          # Test programs
│   ├── test_parser.c
│   └── test_executor.c
├── scripts/        # Helper scripts
│   └── run.sh
└── assets/         # Images/screenshots
```

└── architecture.png

Note: For creating this structure use mkdir and touch commands.

2. To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing.

Task-1: Using a manual command understand the fork() and exec() system calls

```
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ ./Practical
This is the Parent Process.
Parent PID = 1337
Child PID = 1338
This is the Child Process.
Child PID = 1338
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ gcc Practical.c
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ cat practical.c
cat: practical.c: No such file or directory
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ cat Practical.c
#include <stdio.h>
#include <unistd.h>

int main() {

    pid_t pid;

    pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
    }
    else if (pid == 0) {
        printf("This is the Child Process.\n");
        printf("Child PID = %d\n", getpid());
    }
    else {
        printf("This is the Parent Process.\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }

    return 0;
}
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ whoami
saikrishna
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ date
Sat Aug  8 09:00:02 UTC 2026
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ gcc Practical.c -o Practical
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ ./Practical
This is the Parent Process.
Parent PID = 1456
Child PID = 1457
This is the Child Process.
Child PID = 1457
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ |
```

Task-2: Write a C program to create a new process and accepting the user commands using fork() and exec() system calls.

```
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ cat Practical2.c
```

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    while (1) {
        printf("MyShell> ");
        fflush(stdout);

        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0) {
            break;
        }

        pid_t pid = fork();

        if (pid < 0) {
            perror("fork");
            exit(1);
        }

        else if (pid == 0) {
            printf("Child PID = %d\n", getpid());

            execlp(command, command, NULL);

            perror("exec");
            exit(1);
        }

        else {
            printf("Parent PID = %d\n", getpid());

            wait(NULL);
        }
    }
}
```

```
    }  
}  
    return 0;  
}  
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ hostname  
DESKTOP-LGFVDTJ  
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ username  
username: command not found  
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ whomai  
Command 'whomai' not found, did you mean:  
  command 'whoami' from deb coreutils-from-gnu (0.0.0~ubuntu25)  
  command 'whoami' from deb coreutils-from-uutils (0.0.0~ubuntu25)  
  command 'whoami' from deb coreutils-from-busybox (0.0.0~ubuntu25)  
  command 'whoami' from deb coreutils-from-toybox (0.0.0~ubuntu25)  
Try: sudo apt install <deb name>  
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ whoami  
saikrishna  
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ date  
Sat Aug  8 09:14:40 UTC 2026  
saikrishna@DESKTOP-LGFVDTJ:~/OSSP$ ./Practical2  
MyShell> Is  
Parent PID = 1734  
Child PID = 1735  
exec: No such file or directory  
MyShell> |
```